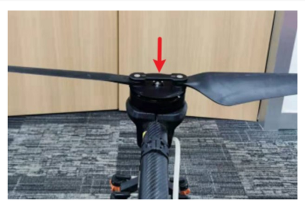
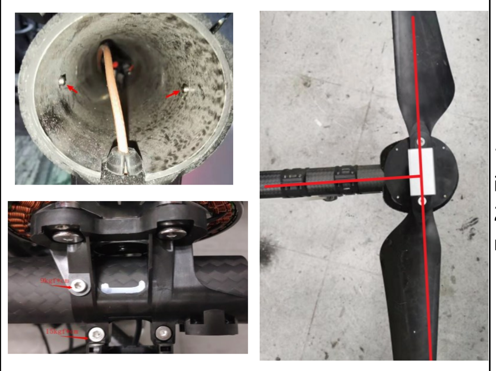
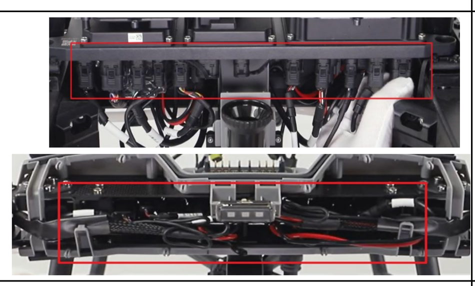
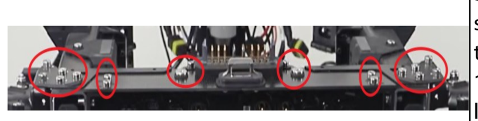
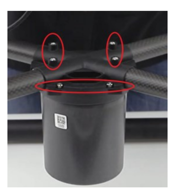
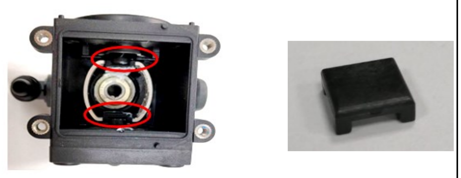
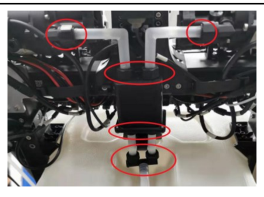
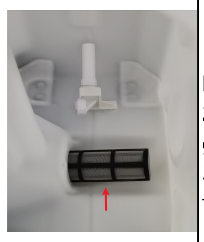
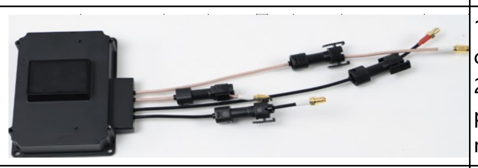
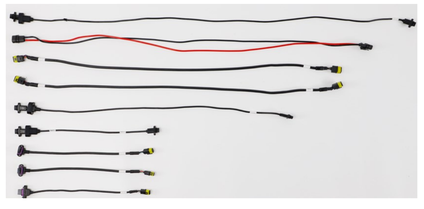
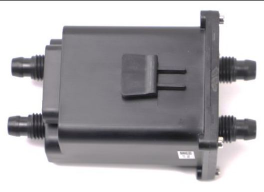
T10 Drone Recommended Maintenance Cycle

Part for service	Module	Recommended service time	Picture	Service item	Service solution	Remarks	Recommended replacement cycle
Propulsion system - motor	Motor	1. Carry out the first inspection for a new drone after 100 flights. 2. Carry out an inspection every 100h after the first inspection. 3. If an ESC is stalling, and the temperature of the motor/ESC is abnormal, carry out this inspection.		1. The upper cover and axis become loose. 2. The axle gear screw becomes loose.	Fix the aircraft arm, and use a tensometer to lift up the propeller adapter to the position as shown by the red arrow (avoid damaging the propeller adapter by the hook). When the tension is 100N, check whether the motor rotator is loose or comes off.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Propulsion system - motor	Motor	1. Carry out the first inspection for a new drone after 100 flights. 2. Carry out an inspection every 100h after the first inspection. 3. If an ESC is stalling, and the temperature of the motor/ESC is abnormal, carry out this inspection.		1. Check whether the internal resistance of the motor is normal. 2. Check whether any cable in the motor is disconnected or whether the terminal is broken.	1. Disconnect the motor and ESC. 2. Use a multimeter or ohmmeter to test the resistance between every two motor terminals. For example, touch terminal A and terminal B respectively with the probe, select an appropriate range, and read the A-B phase resistance. Measure the A-C phase resistance and B-C phase resistance in the same way. The acceptable resistance value is 60±5m Ω.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Propulsion system - motor base	Motor Base	1. Carry out the first inspection for a new drone after 100 flights. 2. Carry out an inspection every 100h after the first inspection. 3. If an ESC is stalling, and the temperature of the motor/ESC is abnormal, carry out this inspection.		1. Check whether the motor angle is normal. 2. Check whether the motor base moves outwards or is tilted.	1. Remove the motor cover. 2. Visually check whether the motor base is tilted, and use the screwdriver to secure the screw securing the motor base to check whether the screw is loose. 3. The torque of the screw shown at the upper left is 9 kgf*cm. 3. The torque of the screw shown at the lower left is 15 kgf*cm. Unfold the propellers and make the aircraft arm and the propeller vertical, and check the motor angle. Motor angle standards: The angle with the circumference of the carbon tube is tilted toward the inside of the center of the aircraft as follows:	Inspection during repair; the end users can inspect by themselves.	Replacement after 1,000h of usage is recommended.
Propulsion system - ESC fixing piece	ESC Fixing Piece	1. Carry out the first inspection for a new drone after 100 flights. 2. Carry out an inspection every 100h after the first inspection.		There are two screws on each side of the ESC, a total of four screws. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Gently shake the ESC to check whether the ESC is loose. Then use the screwdriver to check whether the screws marked in the red circle are loose. If yes, it is recommended to replace the corresponding screw and apply the medium-strength thread glue. The recommended torque is 8.0 kgf*cm. Check whether the fixing hole on the ESC fixing piece cracks. If yes, replace it.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Propulsion system - propellers	Propellers	1. Carry out the first inspection for a new drone after 100 flights. 2. Carry out an inspection every 100h after the first inspection. 3. If an ESC is stalling, and the temperature of the motor/ESC is abnormal, carry out this inspection.		1. Check whether the propeller is misshapen or damaged. 2. Check whether the propeller is loose.	1. Clean the propeller, hold the end of the propeller, lift it up, and check whether its edges show any cracks. 2. Use an M5 screwdriver to tighten the propeller' s securing screws. Recommended torque: 20.0 kgf*cm	Inspection during repair	Replacement after 700h of usage is recommended.
Propulsion system - propeller adapter	Propeller Adapter	Before daily operation preparation		Check whether the propeller adapter is broken.	Rotate the propellers horizontally to the limit position of the propeller adapter and check whether the limit position is misshapen or broken.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Propulsion system - screw bolts of frame arm	Screw Bolts of Frame Arm	Before daily operation preparation		The frame arm has four screw bolts, two screws for front side and two for read side (as shown on the left). Check whether there is a large deflection clearance and great looseness when the frame arm is folded.	When the aircraft arm sleeve is unlocked, fold the frame arm slowly and check whether there is a lot of looseness or too large of a clearance at the folded part of the frame arm. Use a sleeve to tighten the hexagon nut (recommended torque 20.0 kgf*cm). If the tightening force is too small, replace the nut.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Aircraft components - screws	Securing screws of aircraft arm buckle	Every 100h		A total of 10 screws securing the four buckles. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake the front and rear frames, and visually check whether the frames are loose and whether the screws are loose or broken. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 10.0 kgf*cm	Inspection during repair	Replacement after 1,000h of usage is recommended.
Aircraft components - connector	Aircraft Connector	Every 100h		Check whether the connectors of the power distribution board and cable distribution board are loose.	Slightly pull down the connector, and visually check whether the connector is loose. If yes, mount the connector in place.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Landing gear components - screws	Screws Securing the Middle Frame	Every 100h		There are 11 screws at the each right angle of the middle frame fixing screws, a total of 44 at four right angles. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake and pull the middle frame, and visually check whether the middle frame is displaced or loose. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 15.0 kgf*cm	Inspection during repair	Replacement after 1,000h of usage is recommended.
Landing gear components - screws	Screws Securing the Rear Frame	Every 100h		Screws securing the rear frame as shown on the left: There are a total of 18 screws. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake and pull the rear frame, and visually check whether the rear fram fixing piece is displaced or loose. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 9.0 kgf*cm		
Landing gear components - screws	Screw Bolts Securing the Landing Gear	Every 100h		The landing gear has a total of 20 screws, 10 secures for each side. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake and pull the landing gear, and visually check whether the landing gear is displaced or loose. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 20.0 kgf*cm	Inspection during repair	Replacement after 1,000h of usage is recommended.
Landing gear components - screws	Landing Gear Screws	Every 100h		There are a total of 24 screws securing the bracket. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake and pull the bracket, and visually check whether the bracket is displaced or loose. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 8.0 kgf*cm	Inspection during repair	Replacement after 1,000h of usage is recommended.
Landing gear components - fixing bracket	Radar Fixing Bracket	Every 100h		There are a total of 13 screws securing the radar bracket. Check the screws securing the radar bracket and screws marked in the red circle. 1. Check whether the screw is loose. 2. Check whether the screw is broken.	Slightly shake and pull the radar, and visually check whether the radar is displaced or loose. Use a screwdriver to check whether screws at the marking place are loose. If so, replace them and apply medium strength thread adhesive on the screws. Recommended torque: 8.0 kgf*cm	Inspection during repair	Replacement after 1,000h of usage is recommended.
Spraying System - Delivery Pump	Delivery Pump	Every 100h		The spray flow of the delivery pump becomes small and it cannot be increased by increasing the rotation speed.	Remove the motor of the delivery pump first, and then remove the wear plate indicated by the arrow to check how serious the wear is. The picture on the left is a diagram of the position of the wear plate. The picture on the right is a diagram of a normal wear plate.	Inspection during repair, or inspect when there is an error with the spray flow.	Replacement after 300h of usage is recommended.
Spraying System - Delivery Pump	Delivery Pump	Every 100h		Check whether the delivery pump is water damage.	Start the delivery pump, set the normal spray speed, and visually check whether there is water leakage around the delivery pump. If there is water seepage around the pump cover, remove the screws on the pump cover and replace the delivery pump diaphragm. The picture on the left is a diagram of the position of the diaphragm. The picture on the right is a diagram of a normal diaphragm.	Inspection during repair, or inspect when there is an error with the spray flow.	Replacement after 300h of usage is recommended.

Spraying System - Sprinkler	Sprinkler	Before daily operation preparation		1. Check whether the sprinkler is blocked. 2. Check whether the sprinkler leaks water.	Start the spraying system, and observe the atomization effect of the sprinkler. Check whether there is divergence and whether the sprinkler is blocked. If the sprinkler is blocked, remove and clean it to prevent uneven spraying and excess pressure from damaging the spraying system. Check whether the nozzle and exhaust valve leak water. If yes, replace the sealing gaskets at these two positions.	Water leakage is found in the sprinkler.	Replacement after 1,000h of usage is recommended.
Spraying System - Hose Connector	Hose Connector	Before daily operation preparation		Check the sealing of hose connectors as shown in the figure. Poor sealing of hoses will cause air inlet.	Manually check whether the hose connectors are loose, and visually check whether the connectors are damaged. If damaged, replace them immediately, or air will enter the hoses and cause malfunctioning.	Inspection during repair	Replacement after 1,000h of usage is recommended. If the hose connector is misshapen, replace it.
Spraying System - Hose	Hose	Before daily operation preparation		Check whether the hose is worn and broken.	Visual inspection, hand feeling	Inspection during repair	Replacement after 1,000h of usage is recommended. If the sealing piece is leaking water or misshapen, replace it.
Spray Tank/Spraying System	One-Way Valve Sealing	Before daily operation preparation		Check whether the sealing ring of the one-way valve is misshapen or damaged.	Remove the spray tank, and check whether the sealing ring is obviously misshapen and sealing surface is damaged. If yes, please replace it immediately, or air will enter the spray tank and cause malfunctioning.	Inspection during repair	Replacement after 1,000h of usage is recommended. If the sealing piece is leaking water or misshapen, replace it.
Spraying System - filter	Filter	Before daily operation preparation and after the operation	 	1. Check whether the filter is blocked. 2. Check whether the liquid level gauge is dirty. 3. Check whether the connector of the liquid level gauge is loose.	1. Disassemble the upper cover and lower cover of the spray tank, remove the filter and sealing ring, and then clean them. 2. Remove the liquid level gauge and clean it. 3. Check whether the connector of the liquid level gauge is loose. If yes, mount the connector in place.	Inspection during repair	Replacement after 1,000h of usage is recommended. If the sealing piece is leaking water or misshapen, replace it.
Spraying System - Relief Valve	Relief Valve	Before daily operation preparation		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	1. Visually check whether the cable is broken. 2. Power on the aircraft and check the spraying function. Ensure that the air can be discharged.	Inspection during repair	Replacement after 36 months/3,000h of usage is recommended.
Spraying System	Spraying System	Before daily operation preparation and after the operation		Daily cleaning	Clean the spray tank with clean water and start the delivery pump to clean the hoses.	Inspection during repair	Replacement after 1,000h of usage is recommended.
Propulsion system - ESC	ESC	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
RF Module	RF Module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Aerial-Electronics System Module	Aerial-Electronics System Module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Cable Distribution Board Module	Cable distribution board module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Spraying Module	Spraying Module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Radar System - Omnidirectional Digital Radar	Omnidirectional Digital Radar	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Radar System - Upward Radar	Upward Radar	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Power System - Power Distribution Module	Power Distribution Module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Aircraft components - front frame	Front Frame	Every 100h		1. Check whether the front frame is cracked or damaged. 2. Check whether any parts of the structure are loose.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Aircraft components - rear frame	Rear Frame	Every 100h		1. Check whether the front frame is cracked or damaged. 2. Check whether any parts of the structure are loose.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Aircraft components - middle frame	Middle Frame	Every 100h		1. Check whether the front frame is cracked or damaged. 2. Check whether any parts of the structure are loose.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Positioning System - SDR antenna	SDR Antenna	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Positioning System - RTK module	RTK Module	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.

Aircraft cables	Aircraft cables	Every 100h		1. Check whether any cables are damaged. 2. Check whether the cable connector is loose. 3. Check whether the sealing ring on the cable connector is attached in place and whether it is misshapen.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months of usage is recommended.
Spraying system - liquid level gauge	Liquid Level Gauge	Every 3 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Spraying system - flow meter	Flow Meter	Every 6 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Component disassembly and visual inspection	Inspection during repair	Replacement after 36 months/3,000h of usage is recommended.
Aircraft components - battery slider	Battery Slider	Replacement after 2,000 times of plugging and unplugging is required.		Check whether the slider is worn or jammed.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Aircraft components - spray tank slider	Spray Tank Slider	Replacement after 2,000 times of plugging and unplugging is required.		Check whether the slider is worn or jammed.	Component disassembly and visual inspection	Inspection during repair	Replacement after 1,000h of usage is recommended.
Remote controller	Remote controller	Every 3 months		1. Clean it and check whether it is corroded. 2. Check whether it can be powered on and function normally.	Visual Inspection	The service cycle can be adjusted based on actual situations.	Replacement after 36 months of usage is recommended.
Battery	Aircraft Standard Battery	After 100 cycles of charging or 3 months		1. Check whether the battery has swollen or is misshapen. 2. Charge and discharge the battery at least once every 3 months to keep it in good condition.	1. Check whether the battery is broken, swollen, or misshapen more than 2mm. 2. Turn on the battery. There are no warnings shown in the app, and the aircraft can be controlled.	The service cycle can be adjusted based on the actual situation.	Replacement after 1000 cycles
Battery Station	Battery Station	After 200 cycles of charging or 3 months		Check whether the charger has experienced any collision, and whether the charging cable connector is malfunctioning.	Visual Inspection	The service cycle can be adjusted based on the actual situation.	Replacement after 36 months of usage is recommended.